Lecture Notes: How Close Are We to AGI?

A Reality Check for Decision-Makers

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Preface: Who these notes are for

These notes accompany a 90-minute lecture for senior managers at ETH Zurich. The audience is analytically sharp, professionally accomplished, and — by design — *not* specialised in machine learning. You will not be asked to read a gradient or inspect a loss curve. You *will* be asked to read a research paper the way a good board member reads an audit: to see the shape of the argument, the quality of the evidence, and the size of the claim that is actually being made.

The goal is not to tell you when Artificial General Intelligence (AGI) will arrive. Nobody credible knows. The goal is to give you a reference-backed briefing on where the field stands, grounded in six foundational papers, so that you can judge future claims for yourself and ask the right questions of the people who build, buy, or regulate these systems on your behalf.

If you remember only one thing from the whole session, let it be this: *the question is not whether AGI is coming; it is whether your organisation has the evaluation capability to tell, month by month, workflow by workflow, what these systems can and cannot do for you.*

1 What Do We Mean by AGI?

1.1 Why the definition matters — commercially

The phrase "artificial general intelligence" appears in press releases, investor decks, and legal contracts. It is also, as recently as five years ago, a phrase most serious researchers avoided in public. Two things changed. First, GPT-4 was surprising enough that Microsoft Research felt compelled to publish a 155-page technical report titled *Sparks of Artificial General Intelligence* (Bubeck et al., 2023). Second, Google DeepMind — whose explicit corporate mission is to build AGI — decided the community needed a shared vocabulary to argue over. The result was the paper that frames this whole section: *Levels of AGI for Operationalizing Progress on the Path to AGI* (Morris et al., 2024).

For a manager, the definition matters for a concrete reason: procurement contracts, insurance policies, acceptable-use clauses, and regulatory thresholds are starting to invoke "AGI" or "frontier AI" as a trigger. If the term is fuzzy, the obligation is fuzzy. DeepMind's framework is useful precisely because it trades romance for measurability.

1.2 The Morris et al. framework

Morris et al. (2024) observe that prior definitions of AGI — "human-level intelligence," "passing the Turing test," "economically valuable work" — are either unfalsifiable, outdated, or both. They propose a taxonomy organised along two axes:

- **Performance:** how well the system performs relative to humans, ranging from *Level 0* (no AI) through *Emerging* (roughly an unskilled human), *Competent* (50th percentile), *Expert* (90th), *Virtuoso* (99th), up to *Superhuman* (outperforms all humans).
- **Generality:** whether the system is *narrow* (bounded to a specific task or domain) or *general* (capable across a wide range of non-cognitive and cognitive tasks).

That gives a 5×2 matrix. Historical milestones populate it. Deep Blue and AlphaFold sit in the narrow column, as virtuoso or superhuman on their particular task, but zero on generality. Siri and early chatbots were general but emerging at best. The authors argue that today's frontier large language models (they discuss GPT-4, Gemini, Claude) sit in the general column but only at the *Emerging* level: they do many things that an unskilled adult could do, across a wide range of domains, but they do not reliably reach even competent human performance across the board. That is the paper's headline diagnosis, and it is worth stating clearly: *by DeepMind's own*

framework, there is nothing at the 'Competent General AGI' level yet; we are still at 'Emerging General AGI'.

The framework also articulates six design principles that are worth reading even if you never look at the matrix again:

1. Focus on *capabilities*, not on *mechanisms*. What the system can do is measurable; whether it does so via "real" understanding is not.
2. Focus on *generality and performance* separately. Collapsing them hides where progress actually is.
3. Focus on *cognitive* (and optionally metacognitive) tasks, not physical embodiment. An AGI need not have a body.
4. Focus on *potential*, not deployment. What the model could do, not what it is currently permitted to do.
5. Use *ecological validity*: measure on tasks that matter in the real world, not just academic benchmarks.
6. Treat AGI as a *path*, not a single destination. There will be levels, and each level has its own governance question.

1.3 The counterweight: Chollet's critique

No briefing on this topic is honest without the other side. François Chollet's *On the Measure of Intelligence* (Chollet, 2019) argues that benchmark performance — the raw material of the Morris taxonomy — is a misleading indicator. Real intelligence, Chollet argues, is *the efficiency with which a system acquires new skills in a previously unseen domain, using limited prior knowledge and limited experience.* By that measure, a system that scores highly on tasks statistically similar to its training data has demonstrated *memorisation plus interpolation*, not intelligence. Chollet's ARC-AGI benchmark, designed to resist this kind of interpolation, was only substantially cracked by frontier models in late 2024 — and even then at great computational expense.

The responsible reading is not to pick a side. It is to hold both frames at once:

Key insight

Morris et al. tell you *how to locate today's systems on a shared axis.* Chollet tells you *how to test whether the location is real, or whether it reflects training-data overlap.* In commercial settings, you need both: the location to decide scope, and the resistance-to-interpolation test to decide trust.

1.4 Managerial implications of Act 1

Three takeaways you can use on Monday morning:

1. When a vendor says "our system is approaching AGI," the correct first question is: *on which axis?* Performance or generality? Against what reference distribution? Without an answer, the claim is marketing.
2. Today's frontier models are, by the most careful published framework, at the *Emerging General AGI* level. That is both more and less than the public conversation suggests. More: it is genuinely a new category of artefact. Less: it is not yet competent across the board, and the gap between "sometimes surprising" and "reliably useful" is where most enterprise pilot projects still fail.

3. The path is continuous, not discrete. You will not receive a press release announcing AGI. You will, if you are paying attention, notice a series of small capability jumps in workflows you already run.

2 Sparks and Specialist Breakthroughs

2.1 The paper that broke the taboo: *Sparks*

In March 2023, a team of senior Microsoft Research scientists published *Sparks of Artificial General Intelligence: Early experiments with GPT-4* (Bubeck et al., 2023). The paper is 155 pages of qualitative probes: draw a unicorn in TikZ code without seeing an image, solve a novel maths olympiad problem, debug human-written code by inferring intent, write and then critique a stanza in the voice of a specified poet. The authors' conclusion, which caused the paper's shockwave, was that GPT-4 shows "sparks of artificial general intelligence" and that they "could reasonably view it as an early (yet still incomplete) version of an AGI system."

For managers, three things about *Sparks* matter more than the conclusion itself.

The honest caveats inside the paper. The authors note that they worked with an *early, unaligned* version of GPT-4 that had not yet been restricted by post-training safety work. They also note that they could not inspect the training data, so they could not rule out contamination (i.e., the possibility that an apparently novel test question had in fact been seen in some form during training). They selected examples they found interesting — explicit about the cherry-picking, but cherry-picking nonetheless. Their methodology is deliberately qualitative.

The subsequent critiques. The Schaeffer–Miranda–Koyejo paper (Schaeffer et al., 2023), which won a NeurIPS 2023 Outstanding Paper Award, showed that many "emergent" capabilities disappear when you choose a smoother metric. That critique does not demolish *Sparks* — most of the qualitative probes do not depend on a thresholded metric — but it does warn that the *shape* of capability growth is partly a measurement artefact.

What *Sparks* actually established. Even under the most sceptical reading, the paper establishes something real: a single system, trained only on next-token prediction on text, could produce coherent and useful outputs across mathematics, coding, vision (through TikZ and descriptions), theory of mind, and cross-domain analogy, at a level that was not achievable by any previous system. That is a qualitative shift, not a marginal improvement.

2.2 AlphaGeometry: what a specialist system can do

Ten months later, in January 2024, a paper in *Nature* showed the other face of the same revolution: not a generalist with emerging abilities, but a *specialist* that decisively exceeds human experts in a narrow but intellectually demanding domain. Trinh et al. (2024) present AlphaGeometry, a neuro-symbolic system for solving International Mathematical Olympiad (IMO)-level Euclidean geometry problems.

The architecture, in plain language. Olympiad geometry problems typically require a creative insight — for example, introducing a previously unmentioned point, line, or circle — after which a symbolic deduction engine can grind out the rest. AlphaGeometry pairs a neural language model (trained to *propose* such constructions) with a fast symbolic deduction engine (which *verifies* and extends them). The neural component suggests; the symbolic component proves. Crucially, the verifier provides a hard ground-truth signal, so "wrong" proposals are filtered out automatically.

The training data trick. The most provocative aspect of the paper, for managers thinking about the future of training data, is *how the neural component was trained*. Human geometry proofs are scarce and hard to formalise. The authors instead generated *synthetically*: they sampled 100 million random geometric diagrams, ran the symbolic engine to discover what was true about each one, and turned the resulting (theorem, proof) pairs into training data. The model was therefore trained on *zero* human-written proofs.

The headline result. On 30 recent IMO geometry problems, AlphaGeometry solved 25. The average human IMO gold medallist solves 25.9 of the corresponding problems; a silver medallist, 22.9; a bronze medallist, 15.2. AlphaGeometry therefore sits at the gold-medal threshold on this particular problem class.

The follow-up. In July 2024, DeepMind's AlphaProof and AlphaGeometry-2 reached a silver-medal performance on the full IMO (across algebra, combinatorics, number theory, and geometry), getting 28 out of a possible 42 points, one below the gold-medal threshold.

2.3 The pattern

Put *Sparks* and AlphaGeometry side by side and the pattern becomes visible:

Key insight

Whenever a domain has a fast, cheap, automatic verifier — mathematics, code, games with clear win conditions, formal specifications — systems can be trained against that verifier and reach world-expert performance, often without any human demonstrations. Whenever a domain lacks such a verifier — persuasion, taste, contested judgement, open-ended research — progress is slower and harder to measure.

The managerial question that follows is uncomfortable and useful: *which parts of your business have (or could have) a verifier?* Regulatory filings do. Unit tests do. Invoice reconciliation does. Board-level strategic judgement does not. The closer a workflow is to having a verifier, the sooner it will be automatable to expert level.

3 From Chatbots to Agents: Measuring Autonomy

3.1 What we mean by "agent"

A chatbot answers a single question. An *agent* takes a goal, decomposes it into sub-tasks, acts in an environment (files, web, shell, other APIs), observes the result, and iterates. The architectural lineage, if you want to drop the right names, starts with *ReAct* (Yao et al., 2023), which interleaved reasoning traces with tool calls; continues with *Reflexion* (Shinn et al., 2023), which added verbal self-critique; and now includes vendor-deployed "computer-use" systems that drive a browser or a desktop the way a human would, with screenshots as input and mouse/keyboard events as output.

For a manager, the important shift is not the architecture. It is that the unit of work has changed. The last generation of AI produced *one-shot outputs*: a translation, a classification, a completion. The current generation produces *trajectories*: sequences of actions towards a goal, taking minutes, then hours, potentially soon days. This is the transition from *tools* to *workers*, and the speed of that transition is measurable.

3.2 The METR task-horizon methodology

The organisation *Model Evaluation and Threat Research* (METR), which has been running independent evaluations of frontier models since 2023, developed a measurement that is becoming the industry's reference (METR, 2024–2026).

The methodology is simple enough to explain on a whiteboard. Take a portfolio of tasks from software engineering, research, and data analysis. Measure how long a *skilled human professional* needs to complete each one. Then give each frontier model a chance to complete the same tasks end-to-end, and ask: *at what human-time horizon does this model succeed at least 50 % of the time?* That number — the *50 % task horizon* — has a nice property: it grows as the model gets more capable, but it does not saturate in the uninformative way that accuracy-on-a-fixed-benchmark does.

The striking finding is this. Plotted on a log scale against release date, the 50 % task horizon of frontier models has been *doubling roughly every seven months*. In 2022, the reference frontier system could reliably complete human-minutes of work end-to-end. By 2024, it could reliably complete human-tens-of-minutes. By 2026, the horizon is in the low hours. If the trend continues at the same doubling time, a task horizon of one working day arrives around 2027, and one working week around 2029.

3.3 How to read an extrapolation like this

This is the moment in the lecture where managerial instincts should fire in both directions at once.

The trend is real and matters. The measurement is done by an independent third party, on tasks that are not the tasks the models were trained for, with human baselines collected rigorously. It is not a vendor benchmark. The doubling time has been stable enough that it survives the introduction of reasoning models and agentic scaffolding.

Extrapolation is a claim, not a measurement. Every exponential trend in technology eventually bends. Hardware speed-ups slowed as Dennard scaling ended; semiconductor feature sizes did not continue to shrink on Moore's original cadence. The METR curve could bend for several reasons: compute cost, data exhaustion, engineering complexity of long-horizon reliability, or a fundamental capability ceiling we do not yet understand. Conversely, it could accelerate — reasoning models, better tool use, and post-training improvements have all produced step-changes in the past eighteen months.

What the plot is actually evidence for. It is strong evidence that (a) agent capability is moving on a measurable trend, (b) we are not yet at any natural ceiling, and (c) the leading indicator you should watch is the trend itself, updated quarterly, not any single benchmark score. What it is *not* evidence for is a specific date.

Key insight

Managers should treat the METR curve the way they treat a leading economic indicator — informative about direction and magnitude, not a forecast with confidence intervals. Plan for the doubling to continue for roughly another eighteen months; re-plan if it bends.

3.4 Adjacent evidence

Two other benchmarks reinforce the picture. *SWE-bench* (Jimenez et al., 2024) measures the ability of language models to resolve real GitHub issues from open-source repositories by editing

multiple files — a task that, two years ago, was far out of reach. Verified variants that eliminate ambiguous issues are now routinely solved by frontier systems above 50 %. *GAIA* (Mialon et al., 2024) targets general assistance tasks that require multimodal input, tool use, and reasoning; humans score about 92 %, and frontier systems have moved from under 15 % at release to above half by 2026. Both benchmarks are, by construction, more contamination- resistant than multiple-choice tests, and both show the same trend as the METR horizon: from unusable, through useful-with-supervision, to outcompeting the median human junior practitioner on the narrow task.

4 When Will AGI Arrive?

4.1 The largest expert survey ever conducted

Grace et al. (2024) report the results of a January 2023 survey of 2,778 researchers who had published in top AI venues (NeurIPS and ICML). The survey is, to date, the largest of its kind. It asks, in carefully designed ways, when various specific capabilities will arrive, when "High-Level Machine Intelligence" (HLMI) — defined as machines capable of every task better and more cheaply than humans — will arrive, and what the consequences might be.

Three numbers from that survey are worth remembering.

50 % by 2047. The aggregated forecast gives a 50 % probability that HLMI exists by 2047. The 10 % probability date is 2027; the 90 % is after 2100. The spread is enormous. Reasonable researchers disagree by fifty years.

A 13-year shift in one year. The same survey, run in 2022, produced a 50 % date of 2060. In one year, the aggregate moved thirteen years earlier. No methodology change of that magnitude occurred; the respondents updated.

38 % give ≥10 % to extreme outcomes. When asked about the probability that advanced AI leads to outcomes "as bad as human extinction or the permanent disempowerment of humanity," 38 % of respondents gave a probability of at least 10 %. A further substantial fraction gave probabilities above 1 %. This is a minority position, but it is a large minority, and the respondents are the people who build these systems.

4.2 How to read an expert forecast

Three points for the manager in the room.

The shift is the signal. A 13-year jump in one year is not a cleaner forecast replacing a noisier one. It is *the same researchers updating hard on new evidence.* When professionals collectively revise their probabilities that much that fast, the correct Bayesian response is not to fixate on the new point estimate but to widen your own uncertainty and shorten your review cycle.

Aggregates hide the disagreement, and the disagreement is the news. "Fifty percent by 2047" reads like a forecast. It is not. It is a weighted average over a community in which a serious minority expects HLMI before 2030 and another serious minority expects it after 2100. Acting only on the mean is like setting your thermostat to the average of "too hot" and "too cold" and expecting to be comfortable.

The 10 % tail is a decision-theoretic event. A 10 % probability of an extremely bad outcome is the kind of number that, applied to a pharmaceutical trial or a bridge design standard, would stop the programme. If your first instinct on hearing "only 10 %" is to discount it, notice that your instinct is domain-specific. The analogous manager in drug development would not discount it. The question worth sitting with is whether the domain analogy should hold.

4.3 The epistemically honest picture

Key insight

We do not know when AGI arrives. We know that the people best positioned to guess disagree by fifty years, and that the community mean has moved thirteen years earlier in one year. The responsible posture is neither "this is hype; ignore it" nor "we have ten years; panic" but: *build organisational capacity to update quickly when the next strong signal arrives.*

5 Managing the Transition

5.1 The *Science* consensus statement

In May 2024, *Science* published *Managing extreme AI risks amid rapid progress* (Bengio et al., 2024). The signatories include Yoshua Bengio and Geoffrey Hinton (both Turing Award laureates, Hinton also a 2024 Nobel laureate in Physics), Stuart Russell, Andrew Yao, Dawn Song, Pieter Abbeel, Trevor Darrell, Yuval Noah Harari, Daniel Kahneman, and several dozen others. The paper is short, written for a policy audience, and explicit about what the authors are and are not claiming.

They are not claiming that AGI is imminent on a specific timeline. They are claiming that (a) capabilities are advancing fast enough that waiting for clear evidence of risk is not a safe default; (b) the current research ecosystem allocates dramatically more effort to capability than to safety; and (c) governance is lagging the technology in ways that previous regulated industries did not tolerate.

5.2 The four governance asks, plainly stated

The paper makes four asks. They are worth stating in ordinary language, because they are the substance most likely to show up in rules and procurement standards within the next 24 months.

1. **Reorient R&D towards safety.** Devote a serious fraction (the paper suggests roughly a third) of frontier AI research effort to safety and ethical use. Today the fraction is small.
2. **Adaptive governance for frontier systems.** Require pre-deployment capability evaluations, incident reporting, and the ability to pause training or deployment in response to risk findings.
3. **Commensurate risk-management practices.** Hold developers of frontier systems to standards similar to those applied in aviation, pharmaceuticals, and nuclear — including third-party audits and liability for harms.
4. **International coordination.** Avoid a race to the bottom by aligning standards across major jurisdictions, including mechanisms for verification of compliance.

Your view on whether these asks are necessary, sufficient, or politically realistic is not the point. The point is that these are now the reference asks that regulators are converging on. The EU AI

Act, the UK AI Safety Institute's evaluation programme (UK AI Safety Institute, 2024), the Bletchley and Seoul declarations, and US executive action are all legible through this framework.

5.3 A manager's playbook

The honest short list — the Monday-morning checklist — is five items.

1. **Adopt a frontier model in one or two scoped workflows.** Pilot with clear success metrics and a named internal owner. Do not pilot AI as a general-purpose initiative; pilot it as a named productivity intervention on a named workflow.
2. **Build an internal evaluation function.** A small team (initially one person) whose job is to run periodic capability checks against *your* workflows, not academic benchmarks. This is the single highest-leverage investment in this decade.
3. **Track three external signals quarterly.** The METR task-horizon update; the leading third-party agent benchmarks (SWE-bench Verified, GAIA); and the UK AISI / US AISI public findings. These together give you a calibrated, vendor- independent view.
4. **Include AI risk in your standard enterprise risk review cycle.** Treat it the way you would a major regulatory change or a supply-chain dependency — with an owner, a register, and a named escalation path.
5. **Avoid irreversible commitments.** Multi-year exclusive contracts, workflows that cannot be rolled back to a human-in-the-loop version, and vendor lock-in on proprietary evaluation data are all failures of optionality in a period of rapid change.

5.4 What the honest executive posture looks like

Neither "ignore it, this is hype" nor "panic, this is existential" is the responsible posture. Both are abdications of managerial judgement. The responsible posture is:

- *Take out appropriate insurance.* You do not need to believe a bad outcome is likely to justify building the capacity to detect and respond to one.
- *Invest in evaluation, not prediction.* You cannot time the arrival of AGI. You can position your organisation to notice when specific workflows flip from "AI-assisted" to "AI-autonomous."
- *Keep AGI trajectory on the executive agenda.* Do not delegate this entirely to IT or procurement. It is a strategic question that will, within this decade, reshape competitive dynamics in most knowledge-work industries.

Closing: The One Thing To Remember

If you take away nothing else from this session, take away this:

> *The question is not whether AGI is coming. The question is whether your organisation has the evaluation capability to tell — month by month, workflow by workflow — what these systems can and cannot do for you. The companies that build that evaluation capability in the next two years will be the ones that manage the transition well. The companies that outsource the judgement to their vendors, or their regulators, or their consultants, will not.*

The evidence is mixed, the timelines are contested, and the people who build these systems disagree with each other by decades. But on one point they agree: the rate of change has accelerated, and the window for building institutional capacity to respond is now.

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